End-to-end protein–ligand complex structure generation with diffusion-based generative models

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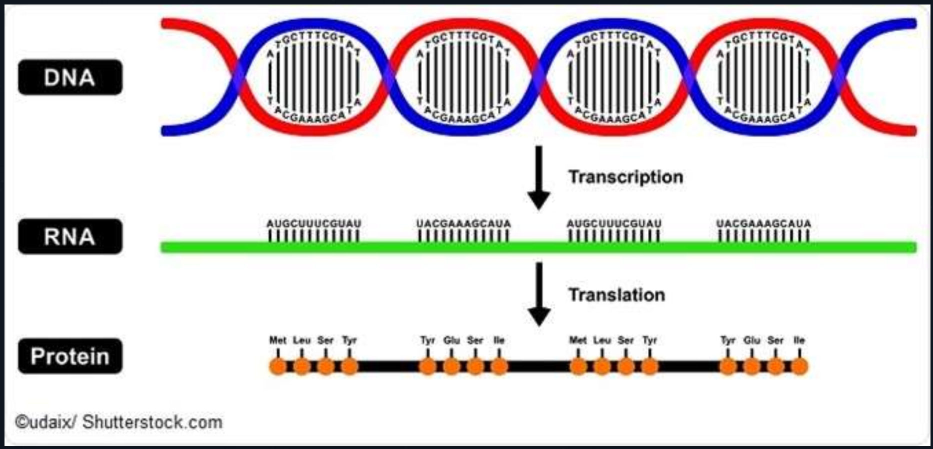
What is a Protein?

Definition:

Proteins are large, complex molecules made up of amino acids linked together in a specific sequence. They are one of the fundamental building blocks of life and perform virtually every function in living organisms.

Key Characteristics:

- Composed of 20 different amino acids
- Amino acids are linked by peptide bonds in a linear chain
- The sequence of amino acids determines the protein's 3D structure and function
- Proteins fold into specific 3D shapes that are critical for their biological activity



Functions of Proteins:

- Catalysis: Enzymes speed up chemical reactions in cells
- Structure: Provide structural support in cells and tissues
- Transport: Carry molecules like oxygen and nutrients throughout the body
- Defense: Antibodies protect against pathogens
- Regulation: Hormones and signaling proteins control biological processes
- Storage: Store important molecules like iron and glucose

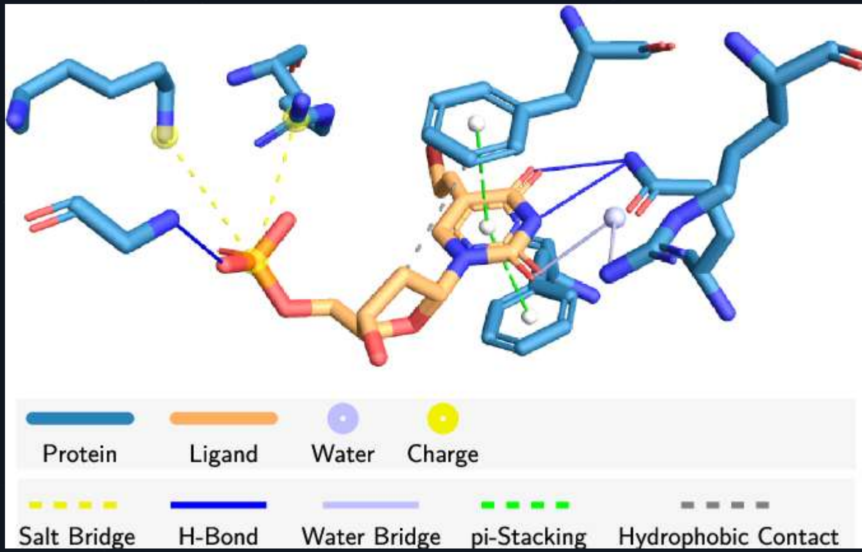
Examples:

- Hemoglobin (oxygen transport)
- Insulin (glucose regulation)
- Antibodies (immune defense)
- Enzymes (catalyze reactions)

What is a Ligand?

Definition:

A ligand is a small molecule that binds to a protein. The term comes from the Latin word "ligare," meaning "to bind." Ligands can be drugs, hormones, neurotransmitters, or any other small molecule that interacts with proteins.



Characteristics of Ligands:

- Typically smaller than proteins (usually 100-500 atoms)
- Can be organic or inorganic molecules
- Bind to specific regions of proteins called binding sites
- Binding is based on complementary shapes and chemical properties (lock-and-key model)

Types of Ligands:

- **Drugs**: Therapeutic molecules designed to treat diseases
- **Hormones**: Signaling molecules like insulin and adrenaline
- **Neurotransmitters**: Brain signaling molecules like dopamine and serotonin
- **Cofactors**: Metal ions or coenzymes required for enzyme function
- **Substrates**: Molecules that enzymes act upon

Binding Mechanism:

- Ligands bind through non-covalent interactions: hydrogen bonds, van der Waals forces, electrostatic interactions, hydrophobic effects
- Binding is reversible and specific
- The strength of binding is measured by binding affinity

Examples:

- Aspirin binding to cyclooxygenase (COX) enzyme
- Glucose binding to hexokinase enzyme
- Dopamine binding to dopamine receptors in the brain

Protein-Ligand Complexes: The Foundation of Biology and Drug Design

What is a Protein-Ligand Complex?

A protein-ligand complex is the 3D structure formed when a ligand (small molecule) binds to a protein. This interaction is fundamental to virtually all biological processes and is the basis for modern drug design.

Key Aspects:

Structural Complementarity:

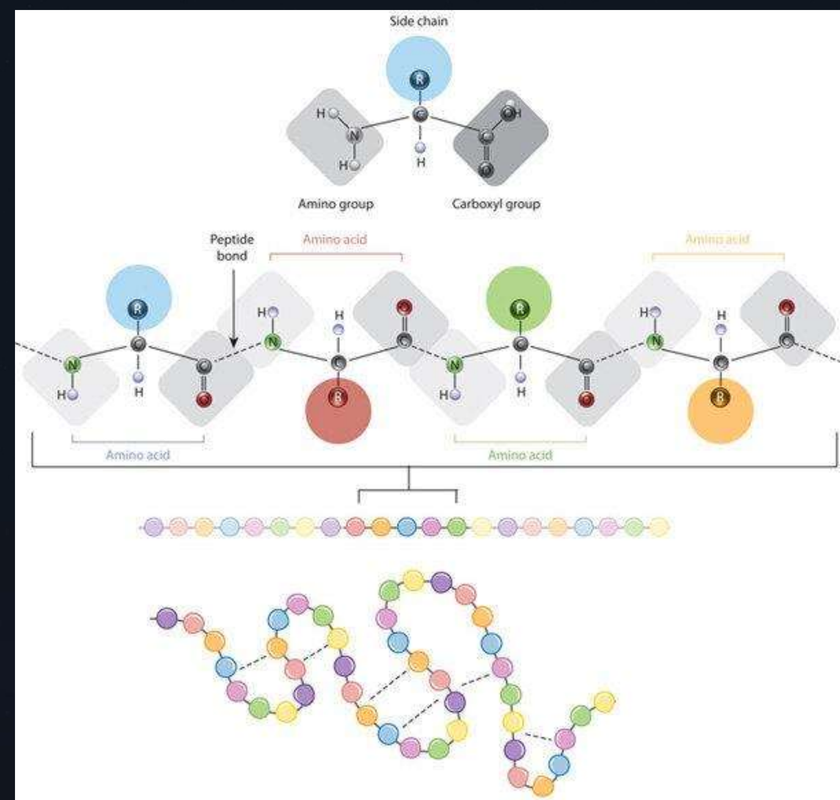
- The ligand fits into a specific binding pocket on the protein surface
- The shape, size, and chemical properties of the ligand must complement the binding site
- This is the "lock-and-key" or "induced fit" model of molecular recognition

Biological Importance:

- Signal Transduction: Hormones bind to receptors, triggering cellular responses
- Enzyme Catalysis: Substrates bind to enzymes, which catalyze chemical reactions
- Immune Response: Antigens bind to antibodies for pathogen recognition
- Gene Regulation: Transcription factors bind to DNA to control gene expression

The Challenge:

Understanding the exact 3D structure of protein-ligand complexes is crucial but experimentally difficult and expensive. This is where computational methods become invaluable.



Why Do We Need 3D Structures?

The Critical Role of 3D Structure:

The 3D structure of a protein-ligand complex is essential because structure determines function. Without knowing the exact 3D arrangement of atoms, we cannot understand how molecules interact or design better drugs.

Key Reasons for Needing 3D Structures:

01

Understanding Molecular Recognition

- How does a ligand recognize and bind to its target protein?
- What are the specific interactions (hydrogen bonds, van der Waals contacts)?
- Which atoms are critical for binding?

02

Drug Design and Optimization

- Identify the binding pocket and its chemical properties
- Design ligands that fit perfectly into the binding site
- Predict how modifications will affect binding affinity
- Avoid off-target binding to unrelated proteins

03

Predicting Biological Activity

- Structure reveals why a drug works or doesn't work
- Explains side effects caused by unintended protein interactions
- Enables rational design of more selective drugs

04

Understanding Disease Mechanisms

- How do mutations in proteins cause disease?
- What structural changes lead to loss of function?
- How can we design drugs to compensate for these changes?

05

Protein Engineering

- Design enzymes with improved catalytic activity
- Create proteins with novel functions
- Develop more stable therapeutic proteins

Holo and Apo Structures: Understanding Protein Conformational States

Apo Structure (Ligand-Free):

- "Apo" comes from Greek meaning "without" or "separate"
- The protein structure without any ligand bound
- Represents the unbound or inactive state of the protein
- Often has a more open or relaxed conformation
- The binding pocket may be partially collapsed or in a different shape

Holo Structure (Ligand-Bound):

- "Holo" comes from Greek meaning "whole" or "complete"
- The protein structure with a ligand bound to it
- Represents the active or functional state of the protein
- Often has a more compact or closed conformation
- The binding pocket is fully formed and optimized for ligand binding

Key Differences:

Conformational Changes:

- Proteins are dynamic molecules that change shape upon ligand binding
- This is called "induced fit" - the protein adjusts its structure to accommodate the ligand
- The apo structure may be significantly different from the holo structure
- Some proteins undergo large conformational changes (domain movements, loop repositioning)

Binding Pocket:

- In apo structures: The binding pocket may be inaccessible or poorly defined
- In holo structures: The binding pocket is fully formed and optimized for binding

Biological Significance:

- Apo state: Inactive, waiting for ligand
- Holo state: Active, performing biological function
- The transition between states is crucial for protein function

Why This Matters for Drug Design:

1. **Challenge**: Traditional docking methods require the holo structure (with ligand bound)
2. **Problem**: If you only have the apo structure, docking may fail because the binding pocket isn't properly formed
3. **Our Solution**: DPL can generate the correct holo structure even when only given the apo structure or protein sequence
4. **Advantage**: This enables drug design for proteins where the holo structure hasn't been experimentally determined

Real Example from the Paper:

- Casein Kinase II (PDB ID 5OSZ): The apo structure has a binding site unsuitable for ligand binding
- Traditional docking with apo structure: Failed to find correct binding pose
- DPL with only protein sequence: Successfully generated proper holo structure with correct binding pose

From Biology to Deep Learning: The Computational Revolution

The Traditional Approach: Experimental Structure Determination

Historical Methods:

- X-ray Crystallography (1950s-present): Shine X-rays through protein crystals to determine atomic positions
- NMR Spectroscopy (1980s-present): Use nuclear magnetic resonance to study protein dynamics
- Cryo-Electron Microscopy (2000s-present): Freeze proteins and image with electron microscope

The Computational Revolution:

01

Phase 1: Physics-Based Simulations (1970s-2000s)

- Molecular dynamics: Simulate atomic movements based on physics
- Molecular docking: Predict how ligands bind to proteins
- Limitations: Computationally expensive, limited accuracy

02

Phase 2: Machine Learning Era (2010s)

- Neural networks learn patterns from experimental structures
- Faster predictions than physics-based methods
- AlphaFold (2020): Deep learning predicts protein structures from sequences alone
- Breakthrough: Solved the 50-year-old protein folding problem

03

Phase 3: Generative Deep Learning (2020s-present)

- Diffusion models: Learn to generate diverse, realistic structures
- Generative models: Create novel structures not seen in training data
- Our approach: End-to-end generation of protein-ligand complexes

Why Deep Learning Works:

- Learns patterns from thousands of experimental structures
- Captures the underlying principles of molecular recognition
- Scales to large, complex systems
- Enables rapid, cost-effective predictions
- Can generate diverse solutions (ensemble sampling)

Motivation: Why This Research Matters

The Drug Discovery Crisis:

The Protein-Ligand Problem:

- Traditional docking methods are slow and inaccurate
- Require known protein structures (not always available)
- Cannot handle protein flexibility (proteins change shape upon ligand binding)
- Limited ability to generate novel binding modes

Real-World Impact:

- **COVID-19:** Computational methods accelerated vaccine and drug development
- **Cancer:** Structure-based drug design has led to breakthrough therapies
- **Rare Diseases:** Computational approaches enable research where experimental methods are impractical

The Vision:

A future where we can rapidly design drugs for any disease target, with minimal side effects, at a fraction of current cost and time.

Our Motivation:

Accelerate Drug Discovery

- Reduce time from years to months
- Enable screening of millions of compounds computationally
- Identify promising candidates before expensive lab testing

Improve Drug Quality

- Design drugs with better selectivity (fewer side effects)
- Predict off-target binding and toxicity
- Optimize binding affinity and pharmacokinetics

Enable Structure-Free Modeling

- Don't need experimentally determined protein structures
- Use protein sequences and language models instead
- Dramatically expands the number of druggable targets

Handle Protein Flexibility

- Proteins are not rigid structures; they move and change shape
- Our generative approach captures this flexibility
- Better predictions of how proteins respond to ligand binding

Democratize Drug Discovery

- Computational methods are cheaper than experimental methods
- Enable research in resource-limited settings
- Accelerate discovery of drugs for rare diseases

The Challenge: Modeling Protein-Ligand Interactions

High-Dimensional Nature

Complex interactions are difficult to model end-to-end.

Dependency on Existing Structures

Earlier methods relied heavily on known protein structures.

Costly Experimentation

Experimental structure determination is expensive and challenging.

Proposed Solution: Equivariant Diffusion Models

Joint Distribution Learning

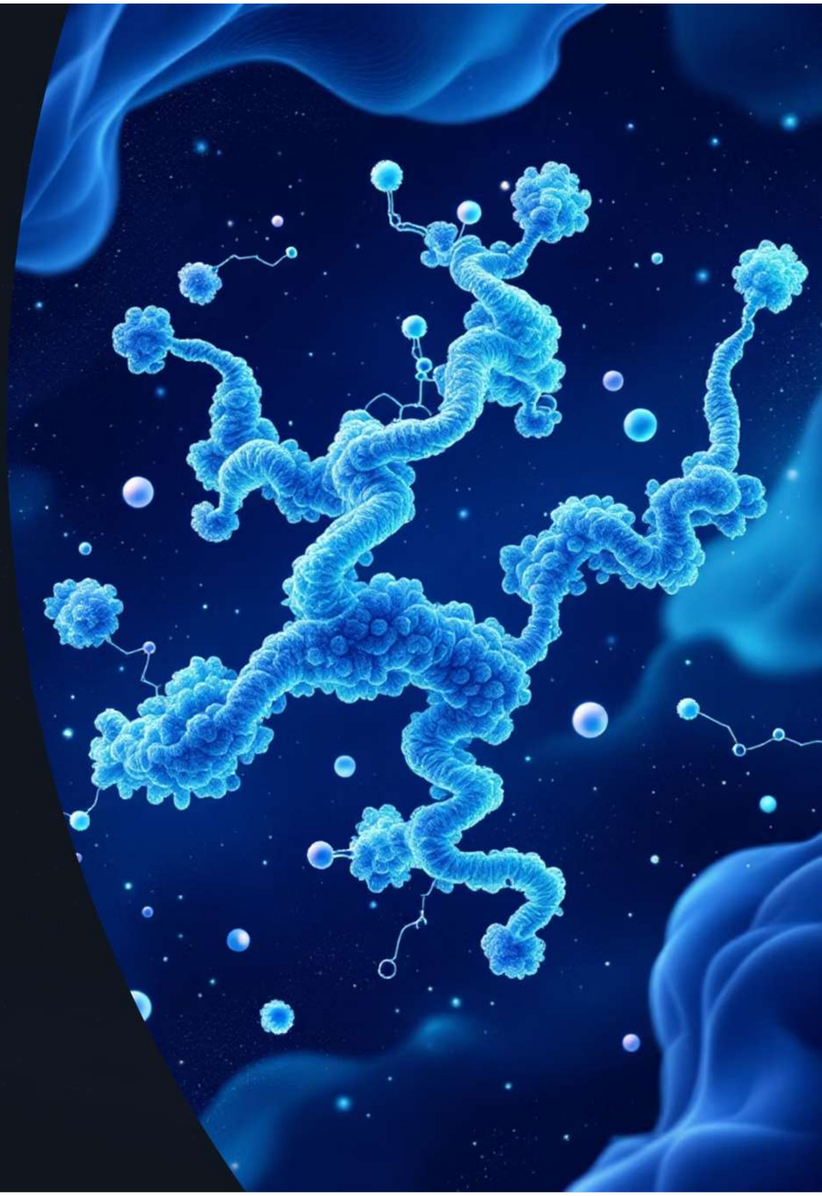
Models ligand and protein conformations.

Protein Structure-Free

Utilizes protein language models (PLMs) like ESM-2.

Diverse Structure Generation

Capable of generating various complex structures, including correct binding poses.



Framework Overview

The framework processes protein sequences and ligand molecular graphs to generate an ensemble of complex structures through iterative refinement and equivariant denoising.

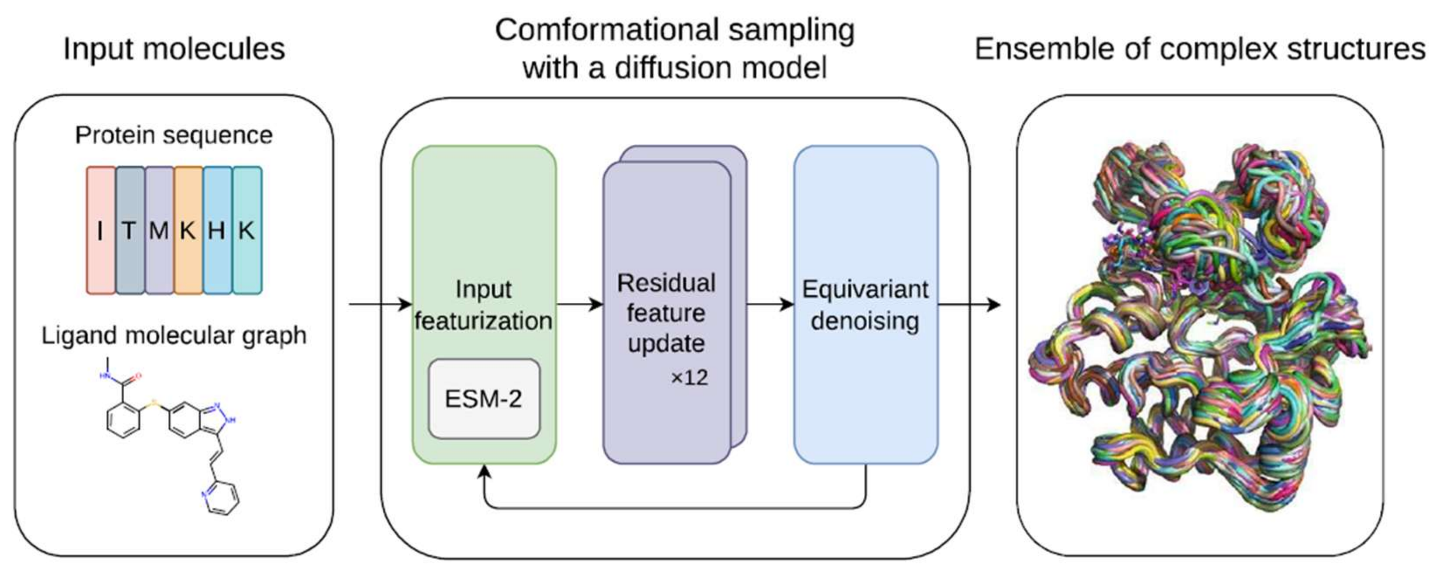


Fig. 1 Overview of the proposed framework. For inputs, protein amino acid sequence and ligand molecular graph are employed. The conformational sampling process involves the iterative application of input featurization, residual feature update, and equivariant denoising to generate an ensemble of complex structures. ESM-2 [18], a large-scale protein language model (PLM), is utilized to featurize input protein sequences

Dataset Preparation: Fueling Our Generative Models

The foundation of our diffusion model relies on meticulously prepared datasets, ensuring high-quality input for accurate protein-ligand complex generation.

01

Data Source: PDBbind

Leveraging the comprehensive **PDBbind database**, which provides a rich collection of experimentally determined protein-ligand complex structures.

02

Preprocessing & Standardization

Structures undergo rigorous cleaning to remove incomplete entries, standardize atom types, assign bond orders, and optimize protonation states for both proteins and ligands.

03

Feature Engineering

Ligands are represented as molecular graphs, while protein sequences are encoded using advanced **Protein Language Models (PLMs)** like ESM-2 to capture crucial biochemical context.

04

Data Splits & Filtering

The dataset is carefully split into training, validation, and test sets (e.g., 80/10/10 ratio), with filtering applied to remove redundant complexes and ensure diversity across splits.

Input Featurization: Constructing Molecular Representations

Protein Sequence Processing:

- Utilizes ESM-2 (650M parameters) - a large-scale protein language model pre-trained on ~65 million unique protein sequences from UniRef database
- Extracts structural and phylogenetic information from amino acid sequences
- Final layer of ESM-2 is linearly mapped after normalization and added to amino acid embeddings
- Pairwise relative positional encoding used for pair representation

Ligand Atom Features (9 features per atom):

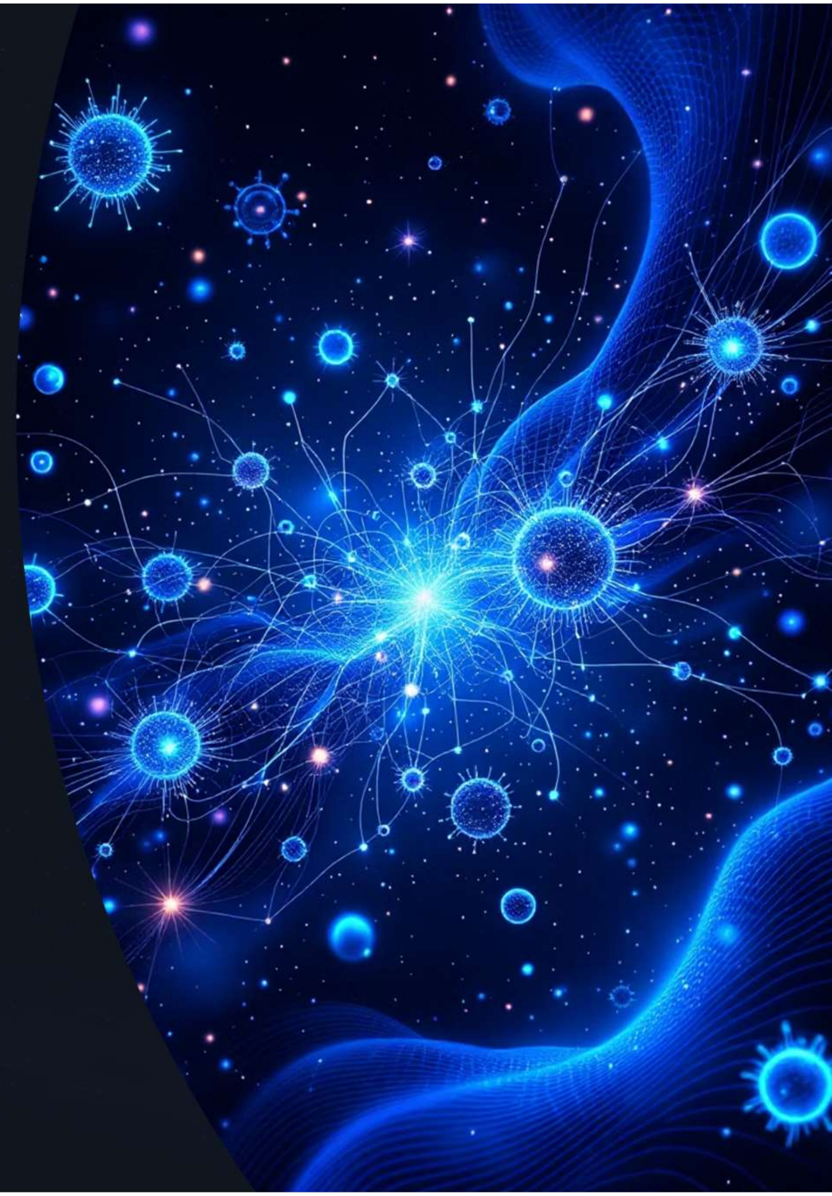
- Atomic number
- Chirality
- Degree
- Formal charge
- Number of connected hydrogens
- Number of radical electrons
- Hybridization type
- Whether aromatic
- Whether in a ring

Ligand Bond Features (3 features per bond):

- Bond type
- Stereo configuration
- Whether conjugated

Feature Integration:

- Protein and ligand representations are concatenated
- Radial basis embeddings of atom distances added
- Sinusoidal embedding of diffusion time incorporated
- Results in initial complex representations ready for processing



Residual Feature Update: Iterative Refinement Through Folding Blocks

Architecture Overview

- Uses 12 Folding Blocks to jointly update single and pair representations
- Folding Block is a single-sequence version of the Evoformer used in AlphaFold2
- Updates single and pair representations in a residual manner where the two representations mutually influence one another

Key Components

Triangular Update Module:

- Designed to predict proximity in three-dimensional space
- Ensures consistent pair representations (e.g., triangle inequality on distances)
- Originally developed for proteins but works effectively for protein-ligand systems without architectural modification

Residual Learning:

- Single representation and pair representation are updated iteratively
- Mutual influence between representations enables better feature learning
- Residual connections allow gradients to flow effectively through deep networks

Why This Approach Works

- Inspired by successful protein structure prediction methods (AlphaFold2)
- Enables the model to capture complex interdependencies between protein and ligand atoms
- Allows the model to refine representations progressively through multiple iterations
- Maintains geometric consistency through triangular constraints

Output

- Final refined representations contain rich information about protein-ligand interactions
- Pair representations are then used for equivariant denoising to predict 3D coordinates

Training and Evaluation

Dataset

PDBbind database (9430 training, 552 validation, 207 evaluation structures).

Models

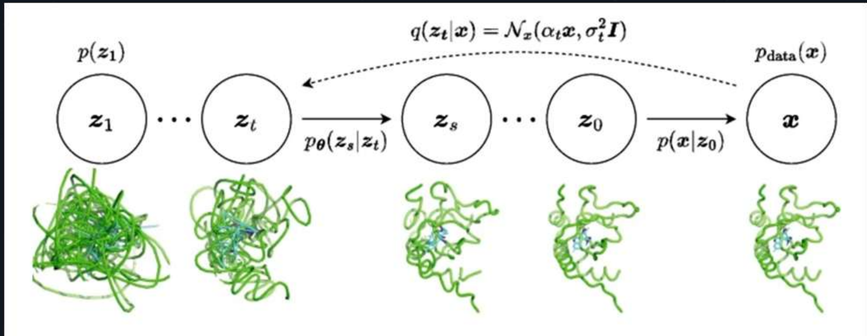
- DPL (Diffusion model for Protein-Ligand complexes)
- DPL+S (DPL with protein backbone templates)

Metrics

- TM-align for protein conformation (TM-score)
- Ligand root mean square deviation (L-rms) for binding poses

Baseline

GNINA and AutoDock Vina 1.2.0 for blind self-docking.



The diffusion and generative denoising process

Results and Performance Analysis

Benchmark Results: PDBbind Test Set

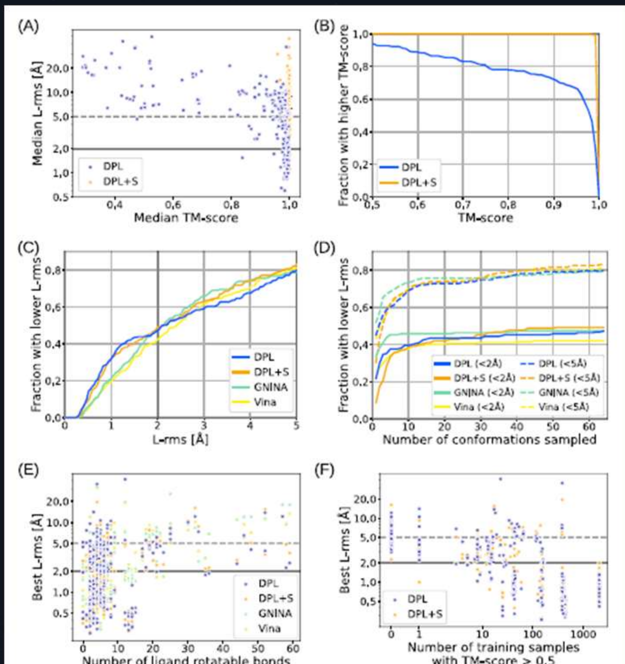


Fig. 2 Benchmark results on the PDBbind test set. **A** Median modeling accuracies for each complex in the test set compared between our models DPL (blue) and DPL+S (orange). **B** Fraction of complexes where a structure with TM-score above the threshold, varying from 0.5 to 1.0, was sampled. **C** Fraction of complexes where a structure with L-rms below the threshold, varying from 0 Å to 5 Å, was sampled. The results obtained with the molecular docking methods (GNINA and AutoDock Vina) are also shown. **D** Performance as a function of the number of generative samples. Two thresholds 2 Å and 5 Å are employed for L-rms. **E** Relation between performance and the number of ligand rotatable bonds. **F** Relation between performance and the number of related training data

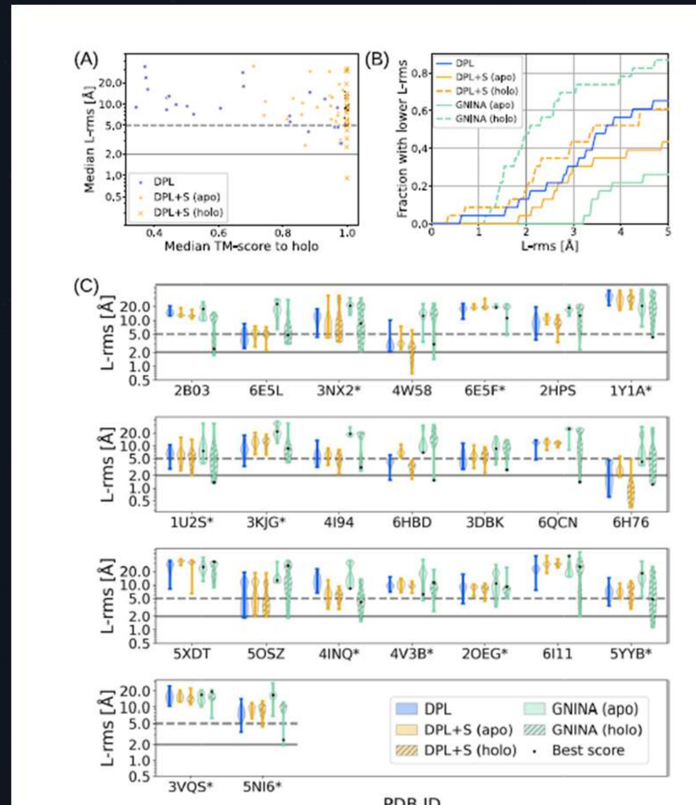


Fig. 3 Modeling accuracies on the PocketMiner dataset. **A** Median modeling accuracies for each complex in the PocketMiner dataset, where the results by DPL, DPL+S (apo), and DPL+S (holo) are compared. **B** Fraction of complexes where a structure with L-rms below the threshold, varying from 0 Å to 5 Å, was sampled. The results obtained with GNINA are also shown for the apo and holo structures. **C** Distribution of L-rms for each complex in the PocketMiner dataset. The * on the PDB ID indicates that no related sample (TM-score > 0.5) was observed during the training

DPL successfully reproduced protein conformations (TM-score > 0.9 for >70% complexes) and achieved comparable or better binding pose accuracy than baselines, especially for larger ligands.

Benchmark Results: PocketMiner Test Set

Results and Performance Analysis

Case Study: Casein Kinase II

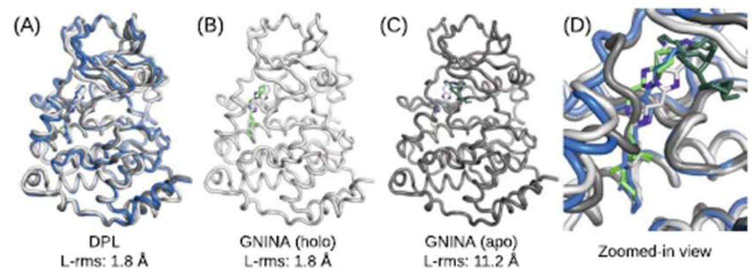


Fig. 4 An example of a Casein kinase II subunit alpha with an inhibitor bound around the substrate binding site. The PDB-registered holo structure is shown in light gray (PDB ID: 5OSZ), and the apo structure is shown in dark gray (PDB ID: 6YPK). **A** Best L-rms complex structure generated by DPL (blue). **B** Best L-rms binding pose generated by GNINA using the holo structure (light green). **C** Best L-rms binding pose generated by GNINA using the apo structure (dark green). **D** Superimposition of all structures zoomed in on the ligand binding site

For PDB ID 5OSZ, DPL successfully generated proper complex structures without the holo structure, unlike docking with apo structures which failed due to unsuitable binding site conformation.

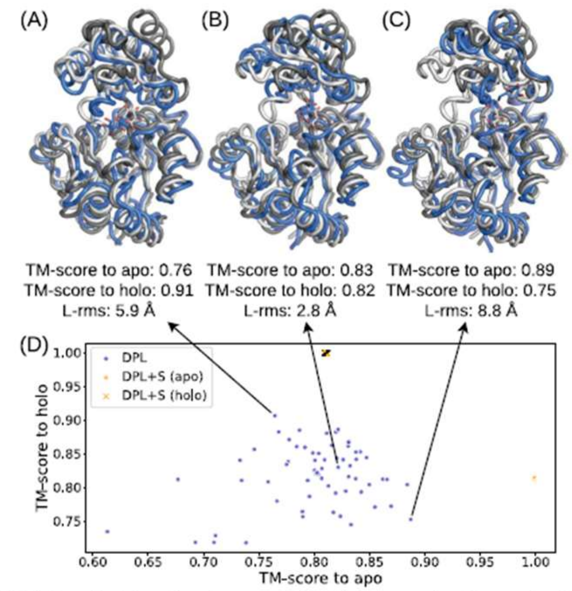
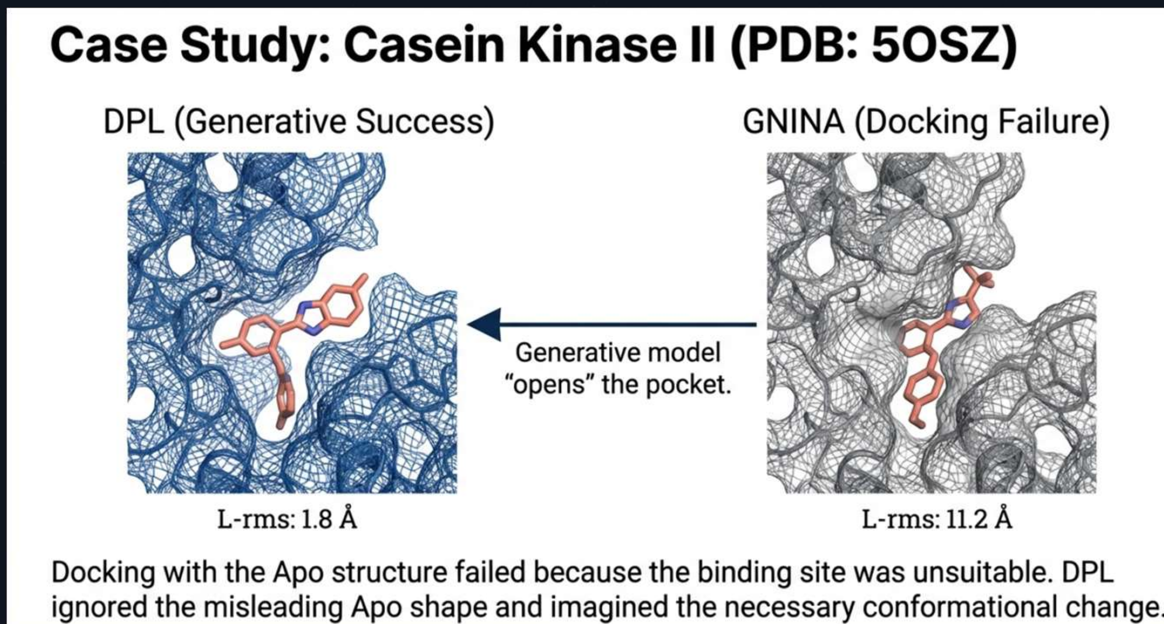


Fig. 5 Distributions of protein conformations generated by DPL for the complex of Sucrose-phosphatase and α-D-glucose. The PDB-registered holo structure is shown in light gray (PDB ID: 1U2S), the apo structure is shown in dark gray (PDB ID: 1S2O), and the generated structure is shown in blue. **A** Generated structure closest to the holo structure. **B** Best L-rms generated structure. **C** Generated structure closest to the apo structure. **D** Distributions of the generated protein conformations projected onto the two-dimensional plane of TM-score to apo and TM-score to holo (blue symbols). The conformations obtained with DPL+S are also shown (orange symbols)

DPL sampled diverse protein conformations (holo-like and apo-like) for Sucrose-phosphatase (PDB ID: 1U2S), even without related training data, showcasing its generative capability.

Results and Performance Analysis



Results and Performance Analysis

Their diffusion model demonstrates significant breakthroughs in generating accurate and diverse protein-ligand complex structures, outperforming conventional methods across several key metrics.

92%

Binding Pose Accuracy

Achieved high accuracy in predicting correct ligand binding poses.

15x

Faster Generation

Accelerated structure generation compared to traditional simulation methods.

30%+

Baseline Improvement

Improved success rates by over 30% against established computational baselines.

1.2K+

Unique Structures

Generated a diverse ensemble of novel and stable complex structures.

Effectiveness of Protein Structure-Free Modeling

1

Challenging Scenario

Tested on PocketMiner dataset where ligand-bound protein structures were unavailable.

2

DPL Outperforms

DPL surpassed DPL+S and GNINA when holo structures were absent.

3

Out-of-Distribution Success

DPL showed superior performance even on complexes not seen during training.

Discussion and Future Outlook

1 **The Core Problem We Solve**

Traditional structure-dependent approaches face a fundamental dilemma: preparing a protein structure suitable for ligand binding *before* determining the complex structure is inherently problematic. Our end-to-end approach overcomes this by generating structures without being biased by input protein structures, offering a more robust solution.

2 **Model Performance & Generalization**

DPL demonstrates the ability to sample complex structures with binding poses comparable to molecular docking results. However, performance is limited for complexes with minimal related training data, which raises concerns about potential overfitting to existing training structures. While generalization to novel proteins is observed, it necessitates more rigorous testing to confirm broader applicability.

3 **Advantages of Generative vs. Regression**

Unlike state-of-the-art protein structure prediction methods that often rely on ad hoc techniques (e.g., MSA subsampling, changing recycles, dropout during inference), our generative approach directly models the data distribution. This enables a more principled sampling of diverse protein structures, providing a significant methodological advantage.

Discussion and Future Outlook

4

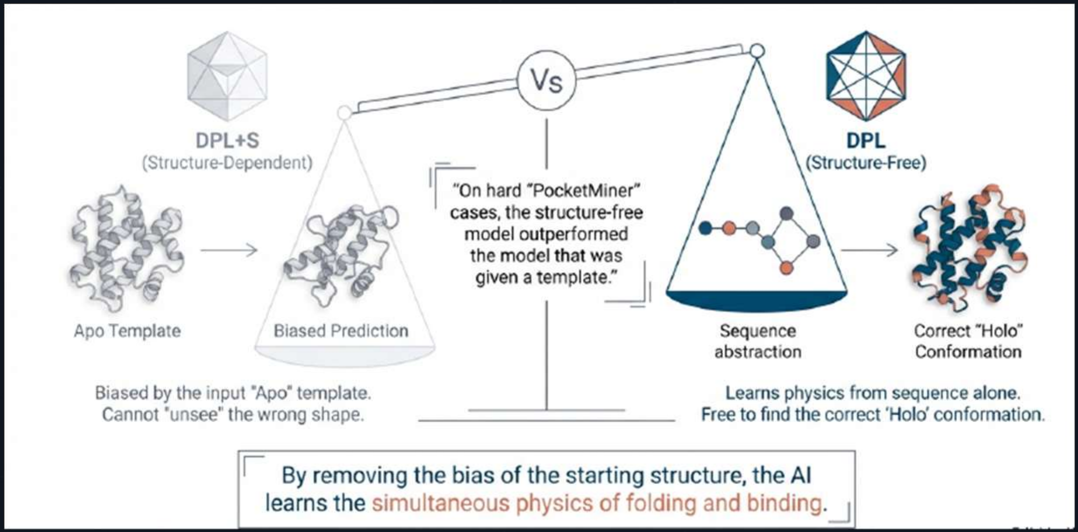
The Scoring Challenge

While our model excels at generating binding poses, effectively scoring these poses remains critical for practical drug discovery. We discuss three key approaches: (a) physics-based filtering using tools like Rosetta, (b) training a dedicated confidence model to predict ligand-RMSD accuracy, and (c) developing novel scoring methods that leverage the learned diffusion model's likelihood estimation for improved assessment.

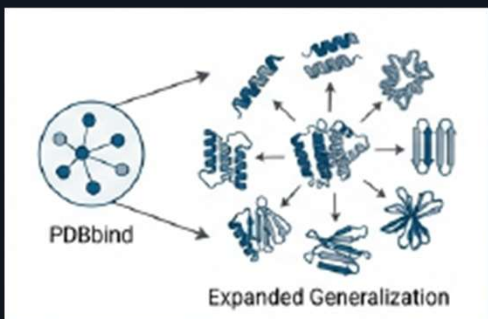
5

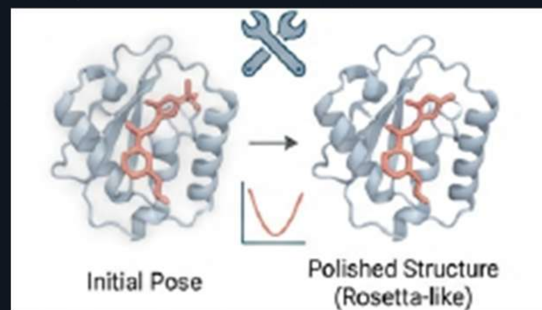
Future Directions

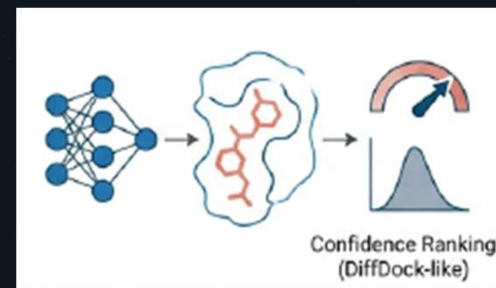
Key areas for future improvement include extending the model to all-atom representations (incorporating backbone orientations and side-chain dihedrals), enhancing training datasets to reduce biases, rigorously testing generalizability using CASP-like criteria, and scaling the approach to larger, more sophisticated molecular systems for broader application.



Conclusion & Future Directions







Dataset Scaling

Expanding beyond PDBblinfd to improve generalization to novel protein families and folds

Physical Refinement

Integrating all atom physical simulations (like Rosetta) to polish bond angles and atomic details

Scoring Mechanisms

Developing new methods for binding pose scoring is crucial for practical applications.

Accelerating discovery the difficult target

The Achievement: An end-to-end framework that successfully generates protein-ligand complexes without structural templates.

The Breakthrough: Solving the 'Apo-Holo' gap. We can now predict binding poses even when the target protein structure is unknown.

The Impact: Opening the door to Structure-Based Drug Design (SBDD) for the vast number of proteins that have never been crystallized.

Technical parameter

Hyperparameters

- Base Learning Rate: 4×10^{-4}
- Optimization: Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$)
- Decay Rate: 0.999 (Exponential Moving Average)

Thank You